Defining performance in computer architecture centers on measuring how quickly and efficiently a computer system executes a given workload. Rather than relying on a single attribute like clock speed, performance is primarily evaluated through time-based metrics and operational throughput.

Key concepts in defining performance include:

- **Response Time (Latency):** The total time required to complete a single task from start to finish, including memory access, input/output operations, and CPU execution time. Decreasing response time directly improves individual task performance.
- **Throughput (Bandwidth):** The total amount of work or number of tasks completed per unit of time (e.g., transactions per second). Increasing throughput improves overall system capacity, particularly in multi-user or parallel environments.
- **CPU Execution Time:** The actual time the processor spends computing instructions for a specific program, excluding time spent waiting for I/O operations or running other background tasks.
- **Relative Performance:** A quantitative comparison between two systems, defined as A and B. If System A completes a task in less time than System B, System A is considered faster by that factor.

Measuring computer architecture performance involves evaluating how quickly and efficiently a system executes programs using measurable metrics rather than raw processor clock speeds.

Key metrics and methodologies used to measure performance include:

- **Execution Time (Response Time):** The total time required to complete a specific task, including memory access, input/output operations, and CPU execution time. It is the most reliable measure of overall system speed.
- **Throughput:** The total amount of work or number of tasks completed per unit of time, which is particularly critical for servers and multiprocessor systems.
- **CPU Execution Time Equation:** CPU Time (or CPU execution time) is the total time a processor actively spends executing instructions for a specific program, excluding time spent waiting for user input, disk operations, or other system processes. It is calculated using the CPU performance equation: Key components affecting CPU Time include:
- **Instruction Count (IC):** The total number of instructions executed, determined by the source code, compiler efficiency, and Instruction Set Architecture (ISA).
- **Clock Cycles Per Instruction (CPI):** The average number of clock cycles required to execute a single instruction, influenced by processor microarchitecture and memory performance.
- **Clock Frequency (f):** The operational speed of the processor hardware, measured in Hertz (Hz).
- **Standardized Benchmarks:** Suites of real-world applications (such as SPEC CPU) executed on different systems to provide unbiased, reproducible performance comparisons.
- **Instruction Throughput Metrics:** Measures like MIPS (Million Instructions Per Second) or MFLOPS (Million Floating-Point Operations Per Second), though they can be misleading because different architectures require different numbers of instructions for the same task.

Ultimately, effective performance measurement depends on evaluating execution time under realistic workloads to determine how hardware changes impact user-perceived speed.